

Regression Analysis for Establishing a Relation Between Weather Parameters

Project Report

Course: Introduction to Data Analytics (IDA)

Group ID: 19

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Abstract

We analyse 1575 consecutive days (2013-01-01 to 2017-04-24) of the Daily Delhi Climate data to establish how mean temperature, humidity, wind speed and mean pressure are related. After repairing 10 physically impossible pressure values and capping 33 implausible wind speeds, we fit, for every one of the six pairs of parameters, a simple linear model and five simple non-linear models (quadratic, cubic, exponential, logarithmic, power). Every estimator, including least squares, Gauss–Newton and the normal equations, is implemented from scratch in base R. Temperature and pressure are by far the most strongly related parameters ($r = -0.881$, linear $R^2 = 0.776$, cubic $R^2 = 0.819$), followed by temperature and humidity ($R^2 = 0.329$); the four relations involving wind speed or humidity–pressure explain less than 20% of the variance. A non-linear curve improves the fit meaningfully ($\Delta R^2 > 0.05$) for one relation only, humidity–pressure ($\Delta R^2 = +0.072$). Removing the annual cycle shows that 76% of the temperature–pressure R^2 is shared seasonality, whereas the temperature–humidity relation is a genuine day-to-day relation (only 8% seasonal).

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1 Introduction and problem statement

A first look at the Delhi data already suggests that the four parameters move together. Averaged over 2013–2017, June is the hottest month (33.7 °C) and also has one of the lowest mean pressures (998.4 hPa), while January is the coldest (13.8 °C) with the highest pressure (1018.1 hPa) and a humidity of 78%; the driest month is May with only 36% humidity. Our task was to turn such impressions into measured relations. The assignment asks us to

- (a) pre-process the data properly,
- (b) find the relation between each pair of weather parameters,
- (c) report the relation using (i) simple linear regression and (ii) simple non-linear regression, and
- (d) calculate R^2 for every case and conclude precisely.

With four parameters there are six pairs. “Simple” regression means one predictor per model, so each relation is studied as $y = f(x)$ for a single pair (x, y) . Because a regression of y on x is not the same as a regression of x on y , we fix a primary direction for each pair (Section 6) and also report the reverse direction.

Two properties of the data shape the whole analysis and are discussed throughout: the observations form a *daily time series* (consecutive days are strongly correlated), and all four parameters follow a pronounced *annual cycle*. A high R^2 between two seasonal variables can therefore reflect their shared seasonality rather than a direct link, and we test this explicitly in Section 8.

2 Dataset description

We use the *Daily Delhi Climate* dataset [1]: daily averages of four weather measurements in New Delhi (Table 1), delivered as two files:

- `DailyDelhiClimateTrain.csv`: 1462 days, 2013-01-01 to 2017-01-01;
- `DailyDelhiClimateTest.csv`: 114 days, 2017-01-01 to 2017-04-24.

Table 1: Variables of the dataset. Units were verified from the observed ranges (e.g. a mean wind speed of about 7 corresponds to 1.9 m/s if read as km/h, which is typical for Delhi; read as m/s it would be implausibly windy every day).

Variable	Meaning	Unit
<code>date</code>	calendar day	—
<code>meantemp</code>	mean temperature of the day	°C
<code>humidity</code>	mean relative humidity	%
<code>wind_speed</code>	mean wind speed	km/h
<code>meanpressure</code>	mean (sea-level) atmospheric pressure	hPa

Data-quality issues. The raw files contain no missing values and no duplicate dates within a file, but:

- **Mean pressure** has 10 physically impossible values, e.g. 7679.33, 1352.62, −3.04 and 12.05 hPa in the training file and 59.0 hPa on 2017-01-01 in the test file. All credible values lie between 991.38 and 1023.00 hPa, and the days around each bad value are normal, so these are recording errors, not weather.

- **Wind speed** has isolated extreme daily means up to 42.22 km/h, against a third quartile of 9.26 km/h.
- **Overlap:** 2017-01-01 appears in both files with different values.

Descriptive statistics of the cleaned data are given in Table 2 and their distributions in Figure 1. Temperature and pressure are bimodal (a cool high-pressure and a hot low-pressure regime), humidity is slightly left-skewed and wind speed is right-skewed.

Table 2: Descriptive statistics of the cleaned daily data ($n = 1575$ days).

Variable	n	Mean	SD	Min	Q1	Median	Q3	Max	Skewness
meantemp	1575	25.23	7.34	6.00	18.52	27.17	31.14	38.71	-0.38
humidity	1575	60.42	16.96	13.43	49.75	62.38	72.12	98.00	-0.34
wind.speed	1575	6.81	4.14	0.00	3.70	6.37	9.26	17.61	0.62
meanpressure	1575	1008.55	7.40	991.38	1001.88	1009.11	1015.16	1023.00	-0.11

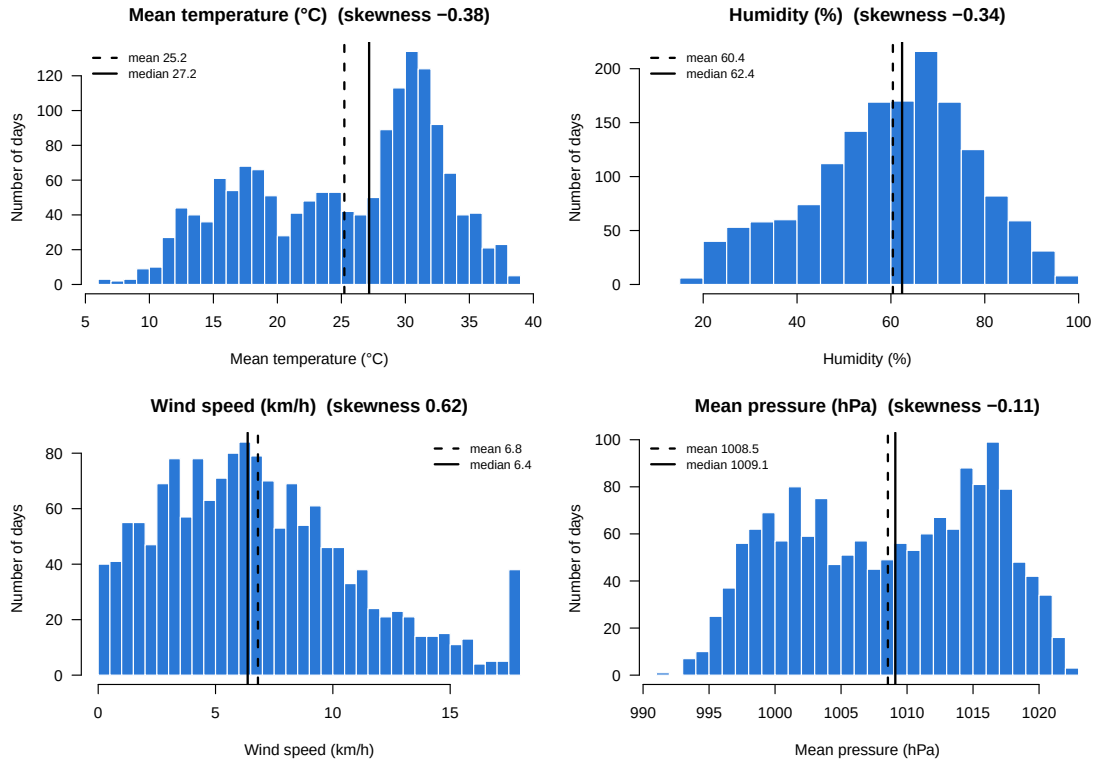


Figure 1: Histograms of the cleaned daily data with mean (dashed) and median (solid).

3 Methodology

All computations were implemented in **base R without any external package**. We did not use any ready-made statistics or modelling function (`lm`, `nls`, `cor`, `var`, `sd`, `quantile`, `median`, `approx`, `solve`, ...); the only built-in statistical function used is `pt()`, the cumulative distribution function of Student's t , for p -values. The submitted script `R/Project_code.R` contains every formula below as a commented function. For our own checking only, we compared all results with R's built-in functions: correlations, linear and polynomial fits agree to better than 10^{-11} , and the Gauss–Newton fits agree with `nls` to within its convergence tolerance.

3.1 Descriptive statistics and correlation

The summary tables are produced by our own mean, variance ($n - 1$ divisor), type-7 quantile [8] and skewness functions. Skewness, $G_1 = \frac{\sqrt{n(n-1)}}{n-2} m_3/m_2^{3/2}$ with $m_k = \frac{1}{n} \sum (x_i - \bar{x})^k$, was the number that told us how much the wind-speed outliers distorted that variable (1.38 before and 0.62 after capping).

For each pair we computed the Pearson coefficient from the centred cross-products,

$$r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}, \quad S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}), \quad S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2,$$

and, as a check for curvature, Spearman's ρ [4]: we rank both variables (ties share the average rank, which matters here because 25 days have zero wind and 33 days share the capped value) and apply the same formula to the ranks. A curved but monotonic relation would make ρ and r differ; we would have flagged a gap above 0.10, and the largest gap we found was 0.027.

With $n = 1575$, the usual test statistic $t = r\sqrt{(n-2)/(1-r^2)}$ (on $n-2$ degrees of freedom) makes every correlation in our data “significant”, so we report it mainly for completeness, together with a 95% interval from Fisher's transformation [5]

$$z = \frac{1}{2} \ln \frac{1+r}{1-r}, \quad \text{SE}(z) = \frac{1}{\sqrt{n-3}}, \quad r \in [\tanh(z - 1.96 \text{ SE}), \tanh(z + 1.96 \text{ SE})].$$

The critical value 1.96 is hard-coded because the assignment rules out `qnorm`.

The more important question for daily weather is how many *independent* observations we really have. Temperature on one day almost repeats the previous day (lag-1 autocorrelation 0.97, pressure 0.97). Following [9, 10] we therefore also use the effective sample size $n_{\text{eff}} = n \frac{1-a_x a_y}{1+a_x a_y}$, where a_x, a_y are the lag-1 autocorrelations of the two variables; for temperature and pressure it shrinks the 1575 days to about 41.

3.2 Simple linear regression

One function, `my_slr`, fits all twelve straight lines (six pairs, two directions). Setting the derivatives of $\sum (y_i - a - b x_i)^2$ to zero gives the closed-form estimates we implemented [2]

$$b = \frac{S_{xy}}{S_{xx}}, \quad a = \bar{y} - b \bar{x}, \quad e_i = y_i - (a + b x_i).$$

From the residuals we obtain $\text{SSE} = \sum e_i^2$, and with $\text{SST} = \sum (y_i - \bar{y})^2$ the two fit measures used throughout the report,

$$R^2 = 1 - \frac{\text{SSE}}{\text{SST}}, \quad R_{\text{adj}}^2 = 1 - (1 - R^2) \frac{n-1}{n-p-1},$$

where p counts the predictor terms (1 for a line, 3 for a cubic). For a single line and $n = 1575$ the adjustment is negligible; it starts to matter only when the polynomials are compared in Section 7. The function also returns RMSE and MAE (in the units of y), the residual standard error $s = \sqrt{\text{SSE}/(n-2)}$, $\text{SE}(b) = s/\sqrt{S_{xx}}$, the t -value of the slope and its 95% interval $b \pm t_{0.975, n-2} \text{SE}(b)$. Because `qt` is not allowed, we find $t_{0.975, n-2}$ by bisection on `pt()`; with infinite degrees of freedom the same routine gives the normal quantiles for our Q-Q plots.

Since the rows are consecutive days, we also summarise the residuals in time order with the Durbin-Watson statistic [6],

$$\text{DW} = \frac{\sum_{t=2}^n (e_t - e_{t-1})^2}{\sum_{t=1}^n e_t^2} \approx 2(1 - \hat{\rho}_1),$$

which is close to 2 for independent residuals and close to 0 when each residual repeats the previous one. Finally, to see whether a handful of days drives a fit, we computed Cook's distance [7] $D_i = \frac{e_i^2}{2s^2} \frac{h_i}{(1-h_i)^2}$ with leverage $h_i = \frac{1}{n} + (x_i - \bar{x})^2 / S_{xx}$ and refitted each line without the days above $4/n = 0.0025$.

3.3 Simple non-linear regression

Five curves with one predictor were fitted:

quadratic	$y = c_0 + c_1 u + c_2 u^2$	
cubic	$y = c_0 + c_1 u + c_2 u^2 + c_3 u^3$	$u = (x - \bar{x})/s_x$
exponential	$y = a e^{bx}$	
logarithmic	$y = a + b \ln x$	$x > 0$
power	$y = a x^b$	$x > 0$

Polynomials are linear in their coefficients. With the design matrix $X = [\mathbf{1}, u, u^2, u^3]$ the least-squares estimate solves the normal equations

$$(X^\top X) \boldsymbol{\beta} = X^\top \mathbf{y},$$

which we solve with our own Gaussian elimination with partial pivoting. The predictor is standardised first: with raw pressures around 1000 hPa the columns of X would be of order 1, 10^3 , 10^6 , 10^9 and $X^\top X$ numerically singular.

Logarithmic model: linear regression of y on $\ln x$.

Exponential and power models are non-linear in b . Start values come from linearising ($\ln y = \ln a + bx$, respectively $\ln y = \ln a + b \ln x$) and fitting by least squares; they are then refined on the *original* scale by Gauss–Newton [3]. With residuals $\mathbf{r} = \mathbf{y} - f(\mathbf{x}; \boldsymbol{\theta})$ and Jacobian $J_{ij} = \partial f(x_i; \boldsymbol{\theta}) / \partial \theta_j$, each step solves

$$(J^\top J) \boldsymbol{\delta} = J^\top \mathbf{r}, \quad \boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \lambda \boldsymbol{\delta},$$

with step halving ($\lambda = 1, \frac{1}{2}, \frac{1}{4}, \dots$) until the SSE decreases. Iteration stops when the relative decrease of the SSE or the relative step is below 10^{-10} (at most 200 iterations); failures are caught and reported, not fatal. For numerical stability the models are fitted in the equivalent centred forms $y = A e^{b(x-c)}$ and $y = A (x/c)^b$ and converted back to a and b . For the exponential model the Jacobian is $\partial f / \partial A = e^{b(x-c)}$, $\partial f / \partial b = A(x-c)e^{b(x-c)}$.

R^2 on the original scale. For *every* model, $R^2 = 1 - \text{SSE}/\text{SST}$ is computed from the residuals of y itself (not of $\ln y$), so all curves are compared on the same footing. Models that are mathematically undefined are skipped and recorded: $\ln x$ does not exist for the 25 days with zero wind speed, so the logarithmic and power models with wind speed as predictor were not fitted.

3.4 Model evaluation and selection

The headline numbers are full-data fits. As an out-of-sample check every model is also fitted on 2013–2016 (1461 days) and evaluated on 2017 (114 days, January–April): $R_{\text{test}}^2 = 1 - \sum (y_t - \hat{y}_t)^2 / \sum (y_t - \bar{y}_{\text{test}})^2$ with $\hat{y}_t$ from the training coefficients. To avoid rewarding a polynomial merely for its extra parameters, the curve selected for each relation maximises

$$\text{score} = \frac{1}{2} (R_{\text{adj}}^2 + R_{\text{test}}^2).$$

An improvement over the line is called *meaningful* if $\Delta R^2 = R_{\text{best non-linear}}^2 - R_{\text{linear}}^2 > 0.05$.

3.5 Seasonality

To separate shared seasonality from day-to-day association we compute *anomalies* $\tilde{x}_t = x_t - \bar{x}_{m(t)}$, where $\bar{x}_m$ is the mean of calendar month m over all years, and correlate the anomalies. The share of the R^2 that disappears, $1 - r_{\text{anom}}^2/r^2$, is the part explained by the common annual cycle. Seasons follow the India Meteorological Department convention: winter (Dec–Feb), summer/pre-monsoon (Mar–May), monsoon (Jun–Sep) and post-monsoon (Oct–Nov).

4 Data pre-processing (2a)

The steps below are performed in Section 2 of the script; Table 3 counts the rows and values affected and Figures 2–3 show the effect.

1. **Combine and sort.** Both files are bound into one data frame with a `source` column, dates are parsed and the rows sorted by date.
2. **Overlapping date.** 2017-01-01 occurs in both files. The training row (10.0 °C, 100% humidity, 0.0 km/h, 1016 hPa) consists of round numbers with saturated humidity and no wind, i.e. a single reading rather than a daily mean; the test row holds proper daily averages but an invalid pressure. We keep the test row and drop the training row, leaving 1575 days.
3. **Date continuity.** The days from 2013-01-01 to 2017-04-24 form a complete daily sequence: no missing and no duplicate dates.
4. **Range checks.** Humidity lies within 0–100% and no wind speed is negative. The 25 days with zero wind are kept: calm days are physically possible.
5. **Invalid pressure.** The 10 values outside 990–1030 hPa are set to missing and filled by linear interpolation in time between the neighbouring valid days. Pressure changes smoothly from day to day (lag-1 autocorrelation 0.97), so interpolation is accurate, and the other three variables of those days are kept. As a cross-check, the interpolated value for 2017-01-01 is 1017.19 hPa, close to the 1016.00 hPa recorded in the dropped training row.
6. **Extreme wind speeds.** Tukey’s rule flags 33 values above $Q_3 + 1.5\text{IQR} = 17.61$ km/h ($Q_1 = 3.70$, $Q_3 = 9.26$), of which 14 also have $|z| > 3$. The largest ones (up to 42.22 km/h) are mostly isolated one-day spikes, up to 17 times the 7-day median, and a 24-hour *mean* of 30–42 km/h would be a strong breeze blowing all day and night. A few others sit in genuine windy spells. Because we cannot prove every high value wrong, the values are *capped* (winsorised) at 17.61 km/h rather than deleted: the day stays in the data and no single point can dominate a least-squares fit. This reduces the skewness of wind speed from 1.38 to 0.62. The IQR rule is preferred to the z -score because mean and standard deviation are themselves inflated by the extremes.
7. **Calendar features.** Month and IMD season are added for the seasonal analysis.

Table 4 shows what the cleaning changed. Only two variables are affected: the standard deviation of pressure drops from 175.3 to 7.40 hPa once the 10 impossible values are replaced, and the capping lowers the maximum wind speed from 42.22 to 17.61 km/h. Temperature and humidity are identical before and after, because no value of theirs was changed.

5 Relation analysis between parameters (2b)

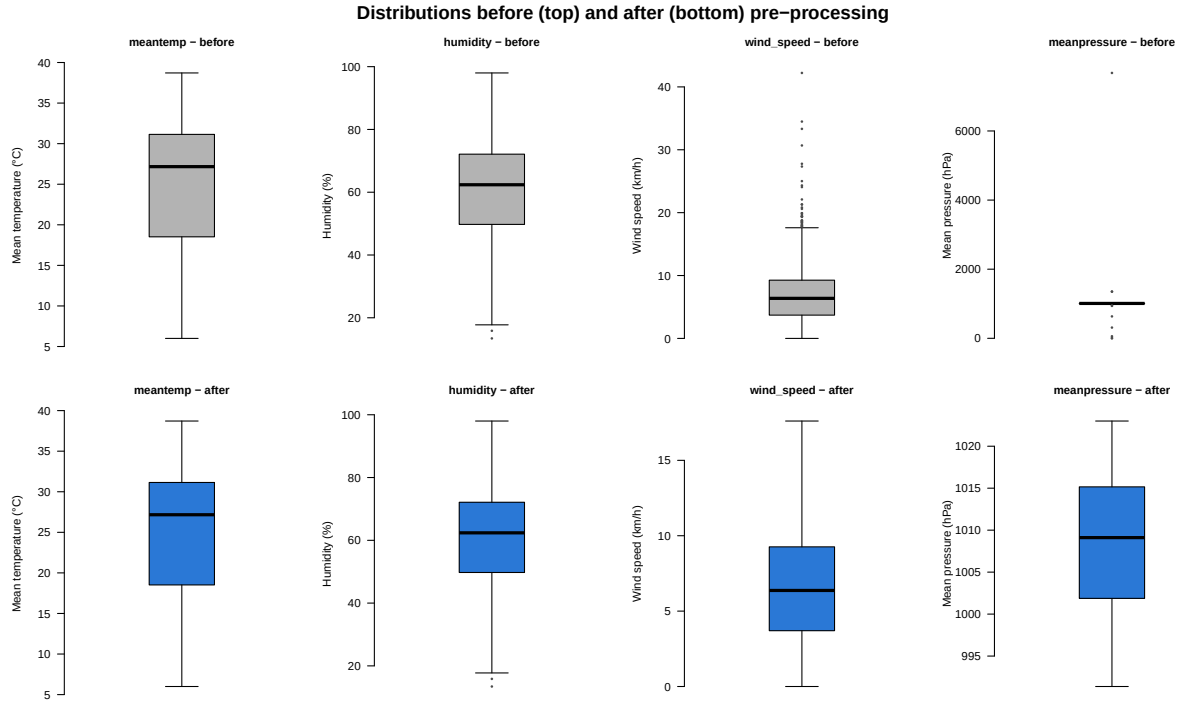
Table 5 ranks the six pairs by $|r|$ and Figure 4 shows both correlation matrices.

Table 3: Pre-processing steps: rows before/after, values corrected per variable and outliers handled.

Step	Check	Variable	Count
Read	Rows in train file	all	1462
Read	Rows in test file	all	114
Combine	Rows after combining	all	1576
Overlap	Overlapping dates removed (train row dropped)	all	1
Date index	Missing dates inserted	all	0
Date index	Duplicate dates	all	0
Range check	Humidity outside 0-100	humidity	0
Range check	Negative wind speed	wind_speed	0
Pressure	Outside 990-1030 hPa, interpolated	meanpressure	10
Wind	Above Q3 + 1.5 IQR, capped	wind_speed	33
Wind	Diagnostic: $ z > 3$	wind_speed	14
Final	Rows after pre-processing	all	1575

Table 4: Descriptive statistics of each variable before and after pre-processing.

Variable	Stage	n	Mean	SD	Min	Q1	Median	Q3	Max	Skewness
meantemp	before	1575	25.23	7.34	6.00	18.52	27.17	31.14	38.71	-0.38
meantemp	after	1575	25.23	7.34	6.00	18.52	27.17	31.14	38.71	-0.38
humidity	before	1575	60.42	16.96	13.43	49.75	62.38	72.12	98.00	-0.34
humidity	after	1575	60.42	16.96	13.43	49.75	62.38	72.12	98.00	-0.34
wind_speed	before	1575	6.90	4.51	0.00	3.70	6.37	9.26	42.22	1.38
wind_speed	after	1575	6.81	4.14	0.00	3.70	6.37	9.26	17.61	0.62
meanpressure	before	1575	1010.59	175.30	-3.04	1001.88	1009.00	1015.18	7679.33	34.64
meanpressure	after	1575	1008.55	7.40	991.38	1001.88	1009.11	1015.16	1023.00	-0.11

**Figure 2:** Distributions before (top) and after (bottom) pre-processing. The raw pressure range is dominated by the invalid values; the wind speed outliers are capped.

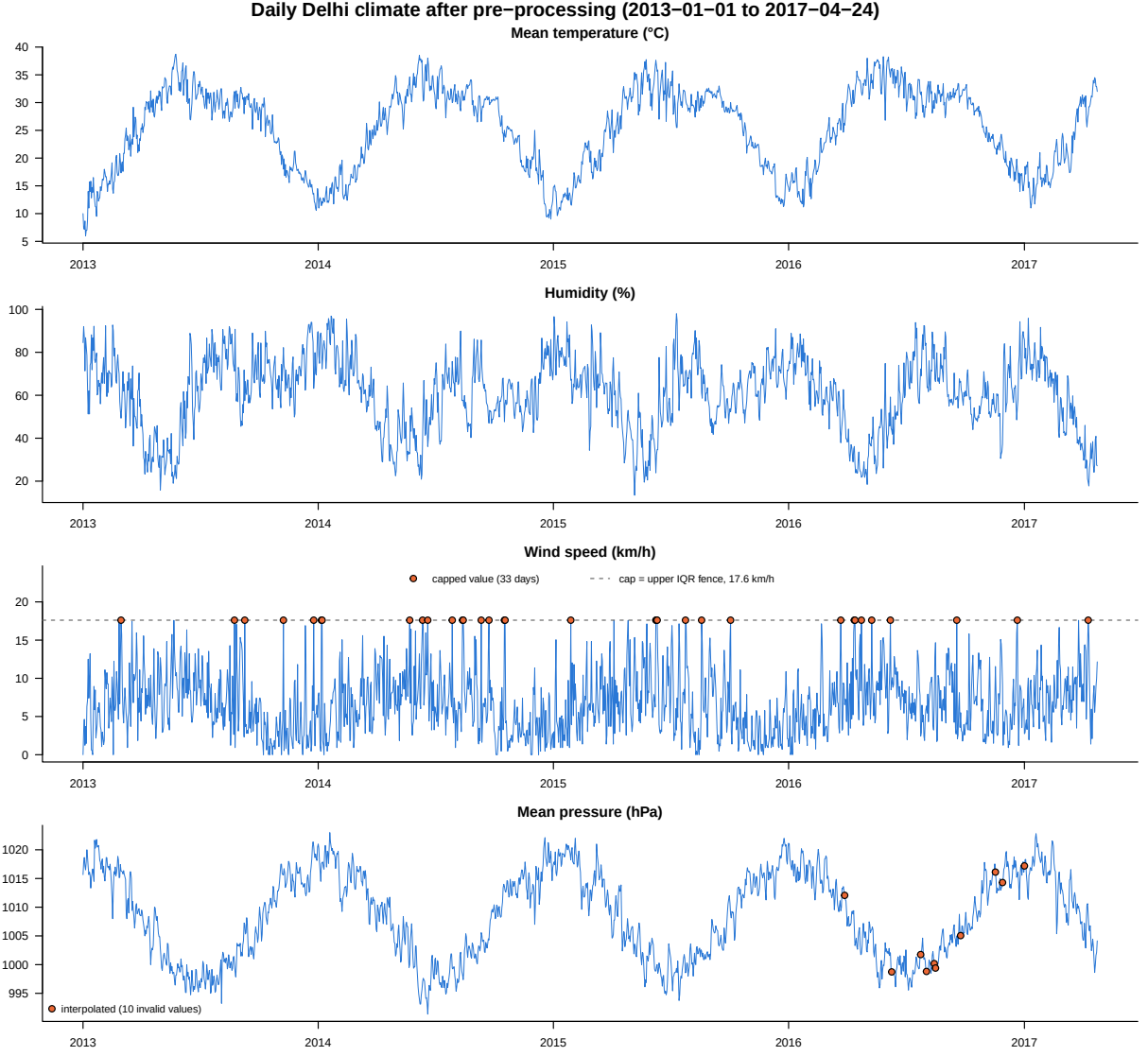


Figure 3: Cleaned daily series. Orange points mark the interpolated pressures (bottom panel) and the capped wind speeds (third panel). All four series show a clear annual cycle.

Table 5: All six pairs ranked by $|r|$: Pearson r with t -test ($t = r\sqrt{(n-2)/(1-r^2)}$, $df = n - 2$) and 95% Fisher- z interval, Spearman ρ , and the p -value recomputed with the autocorrelation-adjusted effective sample size n_{eff} .

Rank	Pair	Pearson r	95% CI	t	p	Spearman ρ	$\rho - r$	n_{eff}	$p(n_{\text{eff}})$
1	meantemp vs meanpressure	-0.881	[-0.891, -0.869]	-73.81	< 0.001	-0.881	0.000	41	< 0.001
2	meantemp vs humidity	-0.574	[-0.606, -0.539]	-27.77	< 0.001	-0.578	-0.004	118	< 0.001
3	humidity vs wind_speed	-0.400	[-0.441, -0.358]	-17.31	< 0.001	-0.413	-0.012	609	< 0.001
4	humidity vs meanpressure	0.339	[0.295, 0.382]	14.30	< 0.001	0.332	-0.008	118	< 0.001
5	meantemp vs wind_speed	0.313	[0.268, 0.357]	13.08	< 0.001	0.340	0.027	543	< 0.001
6	wind_speed vs meanpressure	-0.299	[-0.344, -0.254]	-12.44	< 0.001	-0.315	-0.016	542	< 0.001

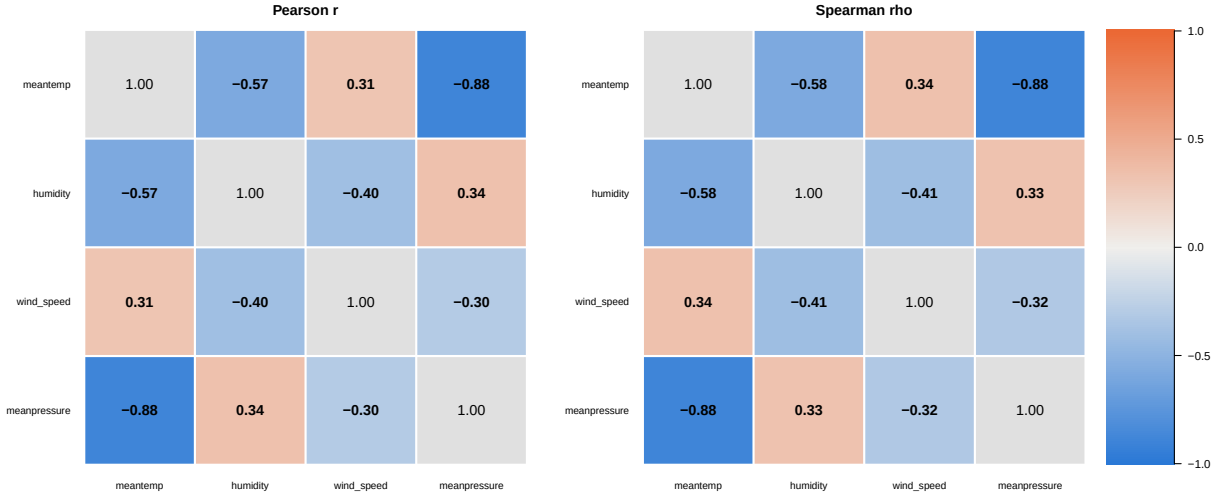


Figure 4: Pearson (left) and Spearman (right) correlation matrices.

Key findings.

- **Temperature and pressure** are very strongly negatively correlated, $r = -0.881$ (95% CI -0.891 to -0.869): hot days have low pressure.
- **Temperature and humidity**: strong negative, $r = -0.574$; hotter days are drier.
- **Humidity and wind speed** ($r = -0.400$), **humidity and pressure** ($r = 0.339$) and **temperature and wind speed** ($r = 0.313$) are moderate; **wind speed and pressure** is weak ($r = -0.299$).
- **Pearson and Spearman agree** for every pair (largest gap 0.027, for temperature–wind speed). There is therefore no sign of a curved *monotonic* relation; both coefficients, however, are blind to non-monotonic patterns, which the scatter plots (Figure 5) do reveal: the temperature–humidity cloud splits into separate bands.
- All correlations are significant ($p < 0.001$), *even after* accounting for autocorrelation: the effective sample size drops to only 41 days for temperature–pressure (lag-1 autocorrelations 0.97 and 0.97), but $|r|$ is so large that the relation remains clearly significant.

Seasonal structure. The calendar month alone explains 87% of the daily variance of temperature, 88% of pressure, 58% of humidity and only 17% of wind speed (Figure 6). Temperature peaks in May–June and pressure is lowest in June–July; humidity has two maxima, in winter and in the monsoon (July–August), and a deep minimum in the dry pre-monsoon months of April–May. The year-by-year distributions (Figure 7) are similar for 2013–2016, so the relations are not driven by one unusual year; 2017 differs only because it covers January–April. This seasonal structure is the key to interpreting the regressions below.

6 Simple linear regression (2c-i)

Direction of each model. Mean temperature is the response for every pair that contains it. For the other pairs we follow the chain from the large-scale state to local weather: pressure drives wind, and both act on humidity, giving `humidity~wind_speed`, `humidity~meanpressure` and `wind_speed~meanpressure`. The direction is a modelling convention, not a causal claim; the

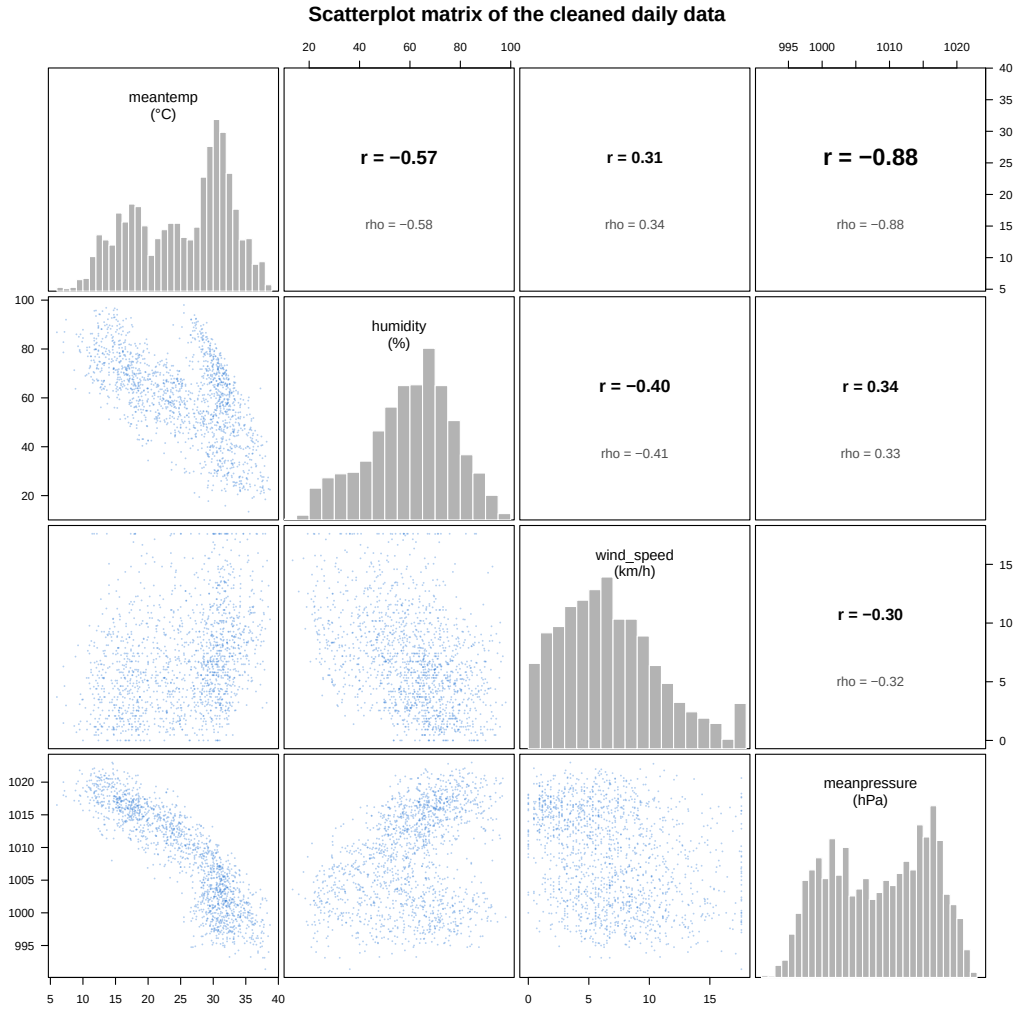


Figure 5: Scatterplot matrix: scatter plots (lower), histograms (diagonal), Pearson r and Spearman ρ (upper).

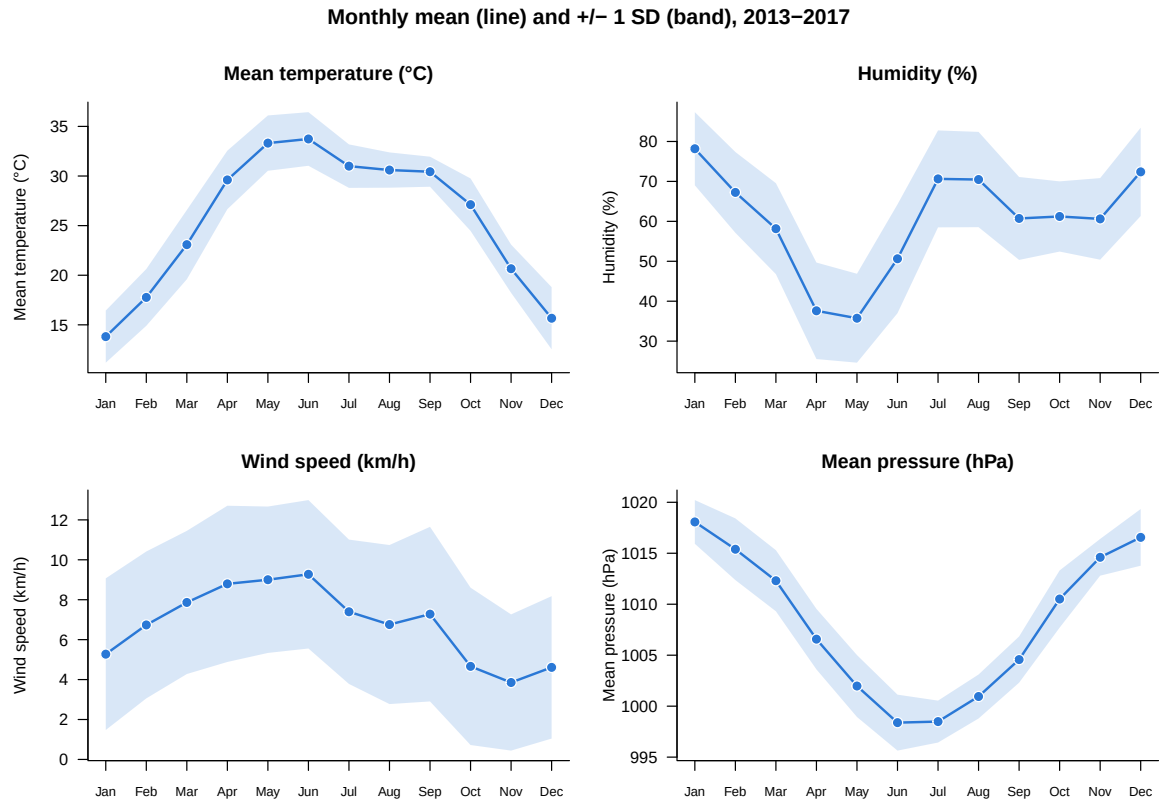


Figure 6: Monthly mean and ± 1 standard deviation of each parameter (2013–2017).

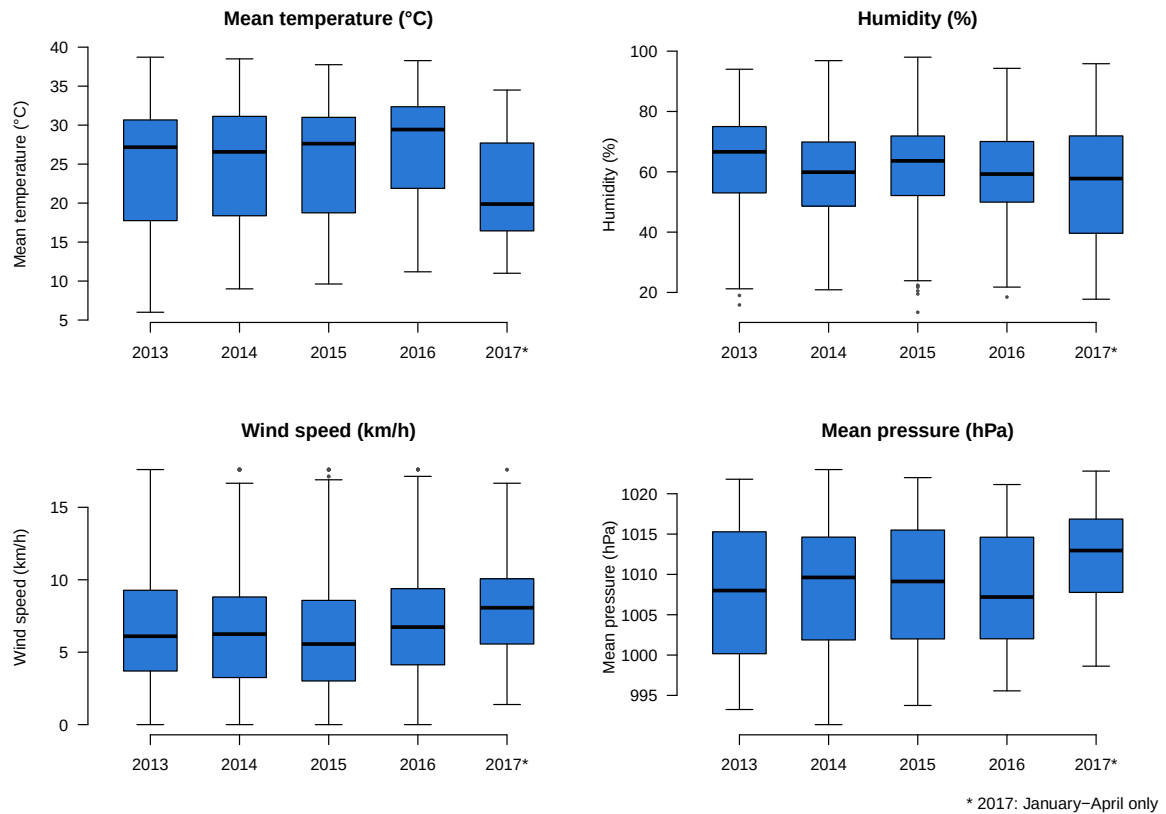


Figure 7: Distribution per year; 2017 covers January–April only, hence its lower temperatures and higher pressures.

reverse fits are reported too. In simple regression $R^2 = r^2$ in both directions, but the slopes are not reciprocals: their product equals r^2 (e.g. $-0.8729 \times -0.8889 = 0.776$ for temperature and pressure).

Table 6: Simple linear regression $y = a + bx$ for every pair, in the primary direction and reversed (rev.). R^2 , adjusted R^2 , RMSE and the Durbin–Watson statistic (DW) refer to the full data (2013–2017); R^2 train/test: fitted on 2013–2016, evaluated on 2017 (Jan–Apr).

Model	a	b	95% CI of b	t(b)	p	R^2	adj. R^2	RMSE	DW	R^2 train	R^2 test
meantemp $\sim$ humidity	40.227	-0.2482	[-0.2657, -0.2307]	-27.77	< 0.001	0.329	0.329	6.009	0.071	0.325	0.122
humidity $\sim$ meantemp (rev.)	93.863	-1.3254	[-1.4191, -1.2318]	-27.77	< 0.001	0.329	0.329	13.885	0.253	0.325	0.310
meantemp $\sim$ wind_speed	21.455	0.5548	[0.4716, 0.6380]	13.08	< 0.001	0.098	0.098	6.966	0.157	0.113	-0.502
wind_speed $\sim$ meantemp (rev.)	2.347	0.1768	[0.1503, 0.2033]	13.08	< 0.001	0.098	0.098	3.932	1.103	0.113	-0.340
meantemp $\sim$ meanpressure	905.615	-0.8729	[-0.8961, -0.8497]	-73.81	< 0.001	0.776	0.776	3.472	0.281	0.773	0.755
meanpressure $\sim$ meantemp (rev.)	1030.977	-0.8889	[-0.9125, -0.8653]	-73.81	< 0.001	0.776	0.776	3.504	0.280	0.773	0.730
humidity $\sim$ wind_speed	71.571	-1.6379	[-1.8234, -1.4523]	-17.31	< 0.001	0.160	0.160	15.535	0.367	0.162	0.099
wind_speed $\sim$ humidity (rev.)	12.713	-0.0977	[-0.1088, -0.0867]	-17.31	< 0.001	0.160	0.160	3.795	1.131	0.162	-0.004
humidity $\sim$ meanpressure	-722.903	0.7767	[0.6701, 0.8832]	14.30	< 0.001	0.115	0.114	15.946	0.258	0.107	0.133
meanpressure $\sim$ humidity (rev.)	999.600	0.1481	[0.1278, 0.1684]	14.30	< 0.001	0.115	0.114	6.963	0.076	0.107	-0.213
wind_speed $\sim$ meanpressure	175.658	-0.1674	[-0.1938, -0.1410]	-12.44	< 0.001	0.090	0.089	3.951	1.095	0.105	-0.356
meanpressure $\sim$ wind_speed (rev.)	1012.191	-0.5350	[-0.6194, -0.4507]	-12.44	< 0.001	0.090	0.089	7.063	0.148	0.105	-0.772

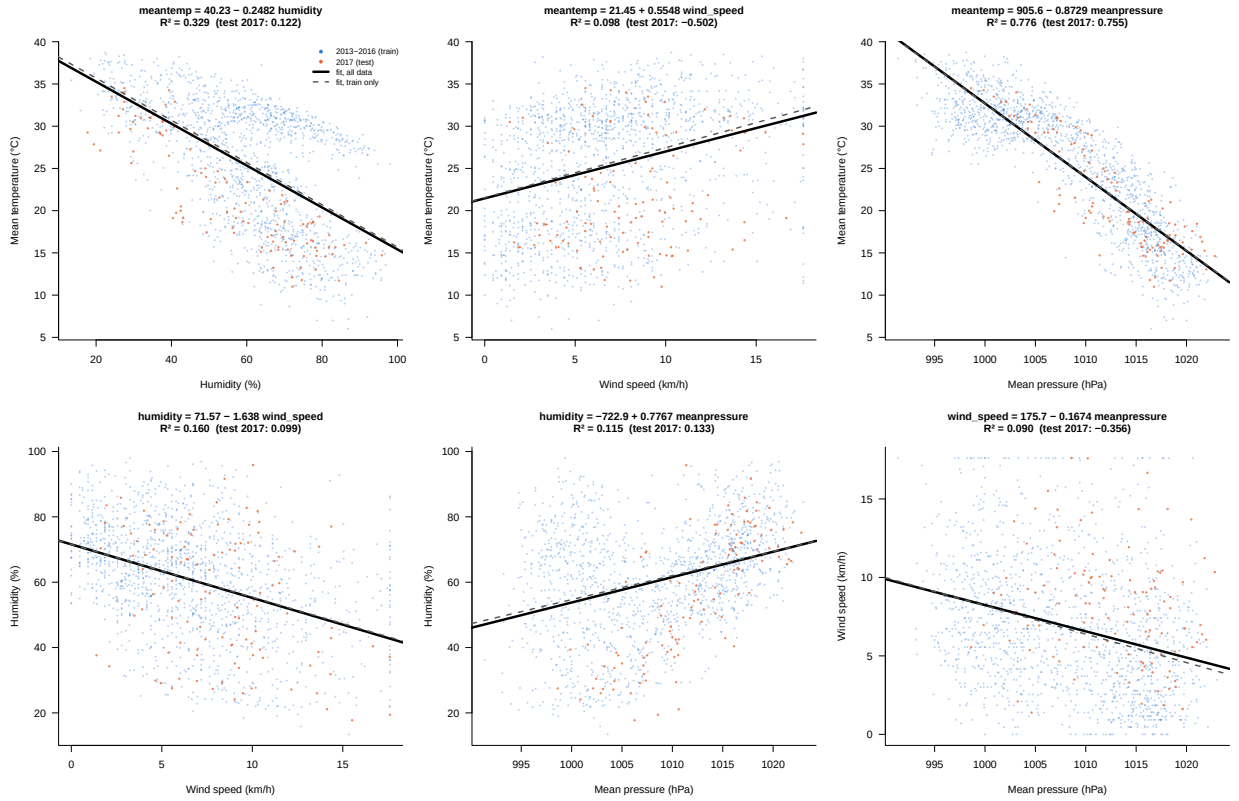


Figure 8: Linear fits on all data (solid) and on 2013–2016 only (dashed); blue: 2013–2016, orange: 2017 test days.

Results. Table 6 lists all twelve linear fits and Figure 8 draws the six primary ones. The fitted temperature–pressure line is $\text{meantemp} = 905.61 - 0.8729 \cdot \text{meanpressure}$: each additional hPa of mean pressure goes with a change of -0.8729°C in mean temperature (95% CI -0.8961 to -0.8497), and pressure alone explains 0.776 of the variance of temperature. The line generalises well: fitted on 2013–2016 it reaches $R^2_{\text{test}} = 0.755$ on 2017. Humidity explains 0.329 of the temperature variance (slope -0.2482°C per % humidity), but only 0.122 on the 2017 test period. The other four linear relations have R^2 between 0.090 and 0.160; for temperature–wind speed and wind speed–pressure the test R^2 is even negative (-0.502 and -0.356), i.e. the lines predict the

January–April 2017 days worse than their mean. The test period covers a single season with a narrower range of values, so a low test R^2 partly reflects that narrower range and not only a poor fit.

Residual diagnostics. Figure 9 shows the residuals against the fitted values with binned means as a simple smoother. For temperature–pressure the binned means trace an inverted U: the line underestimates temperatures in the middle of the range and overestimates them at both ends, a sign of curvature. Humidity–pressure shows the opposite U-shape. For temperature–humidity the residuals split into two bands (monsoon days above, winter days below) rather than a curve. The Q–Q plots (Figure 10) show light tails for temperature–humidity (a mixture of regimes) and right skew for wind speed–pressure. The Durbin–Watson statistics lie between 0.07 and 1.13, far below 2: the residuals are strongly autocorrelated, so the standard errors and p -values above are optimistic. The slopes and R^2 values remain valid as descriptions of the fitted relation.

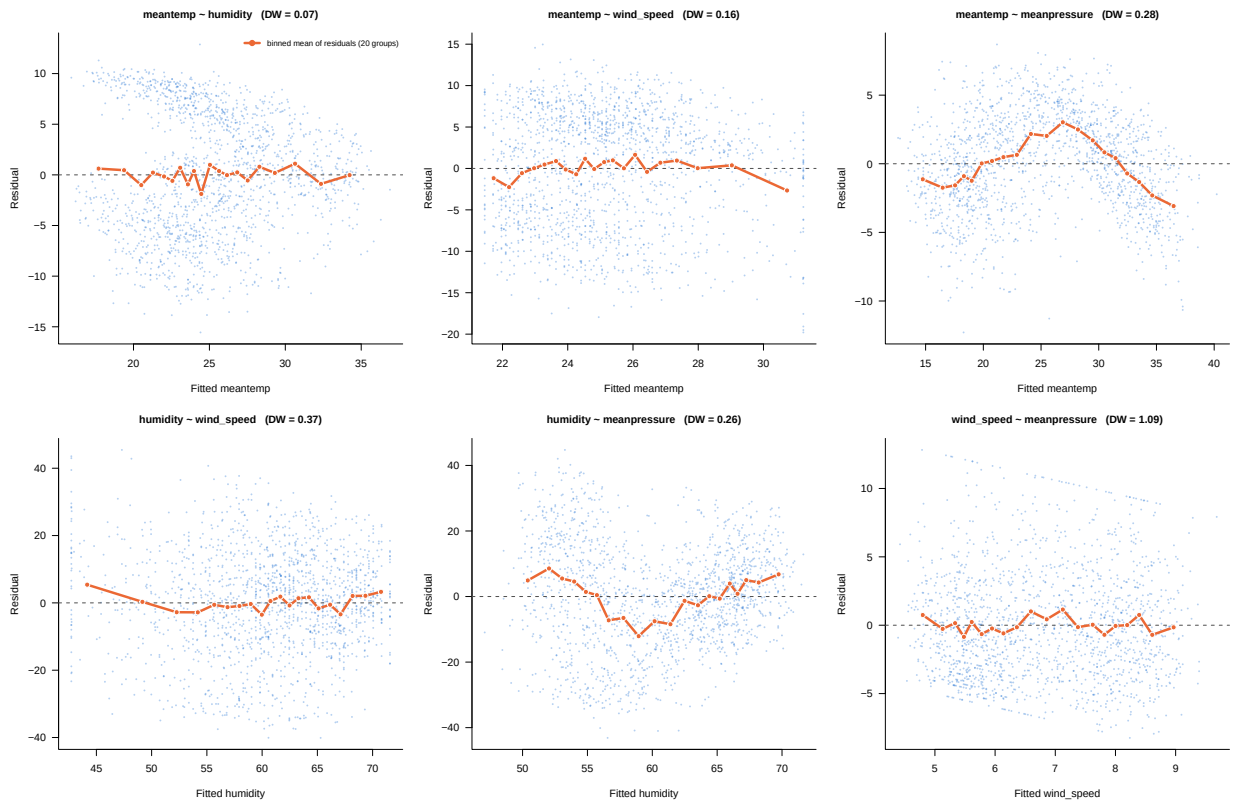


Figure 9: Residuals vs fitted values of the linear models; orange: mean residual in 20 equal-size groups.

7 Simple non-linear regression (2c-ii)

The five curves of Section 3.3 were fitted to the same six pairs and directions (60 non-linear fits, plus the linear reference). All 21 exponential and power fits converged, in 2–6 Gauss–Newton iterations; 6 fits (logarithmic and power with wind speed as predictor) were skipped because $\ln x$ is undefined at zero wind speed. Table 7 lists every curve for the primary directions (the reverse directions are in the CSV file), Table 8 compares the selected curve with the line for all twelve relations, and Figure 11 draws all curves.

Results per relation.

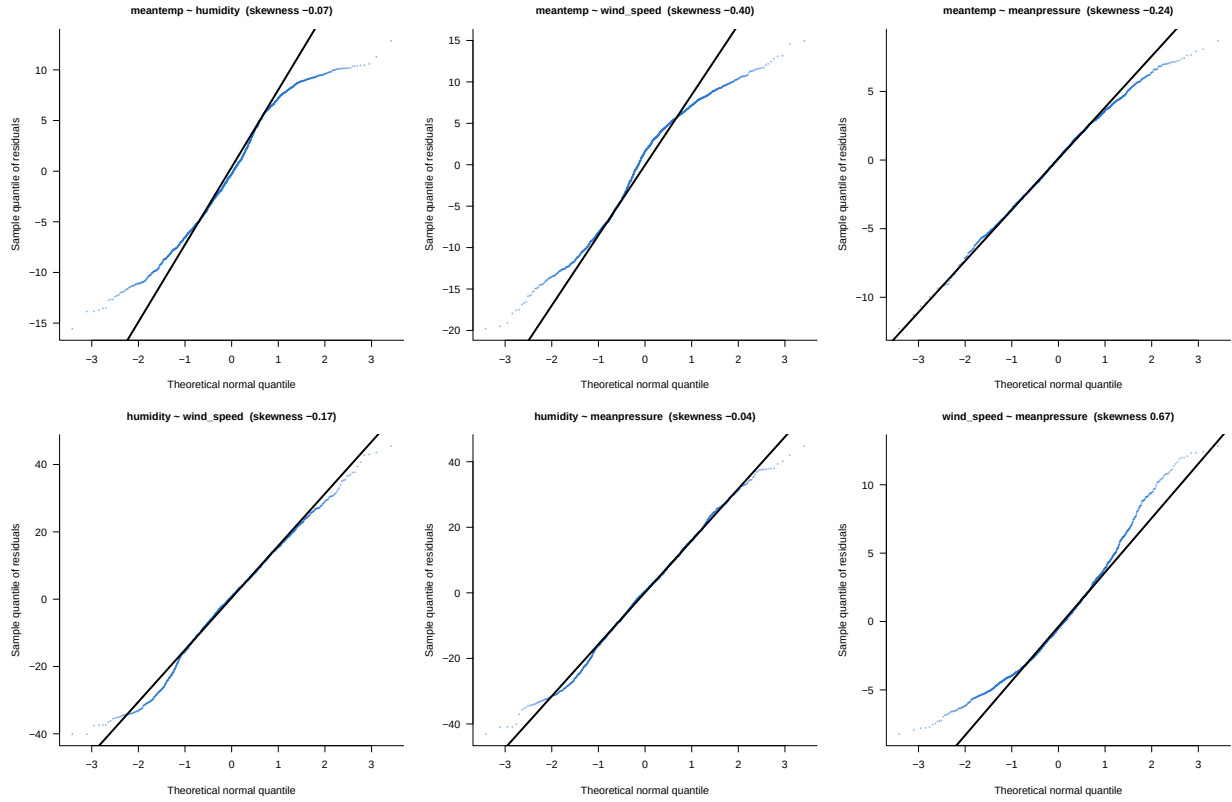


Figure 10: Normal Q-Q plots of the linear residuals.

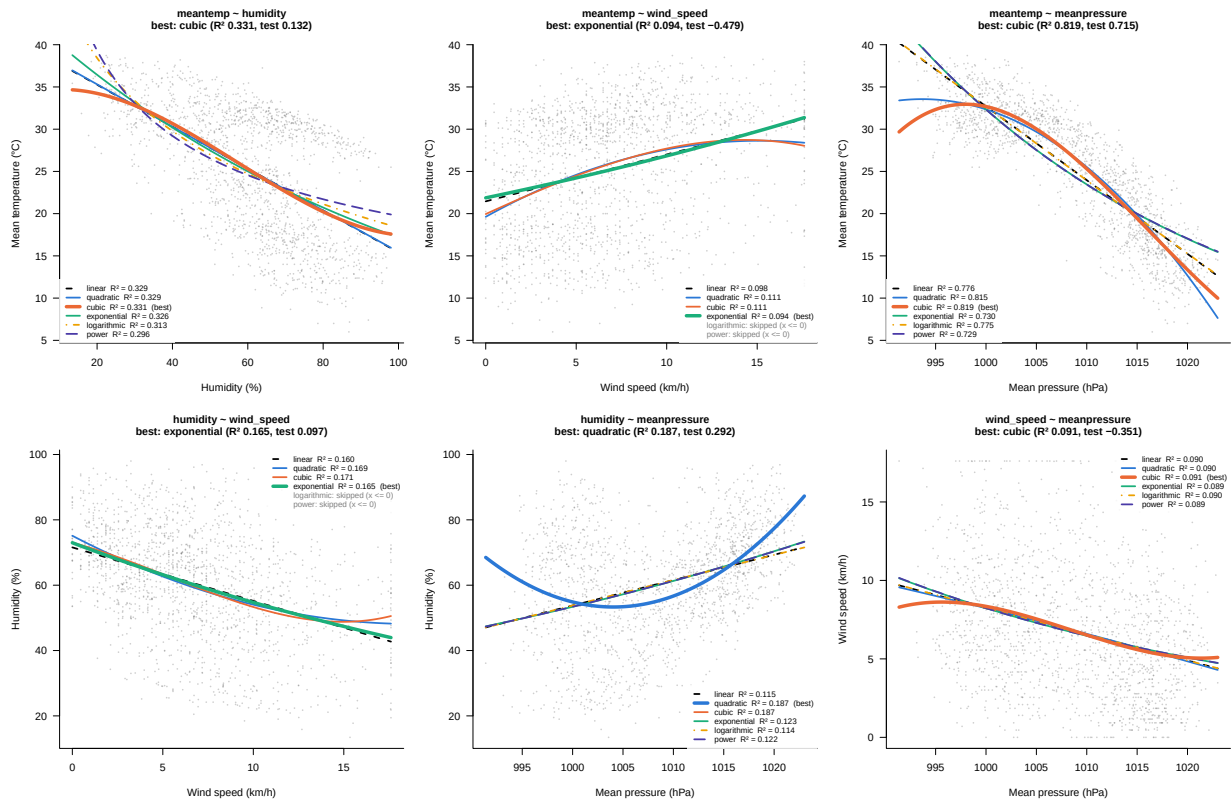


Figure 11: All fitted curves per pair; the selected curve is drawn thick.

Table 7: Linear and non-linear simple regression for the six pairs (primary direction; the reverse direction is in the CSV). R^2 is computed on the original y scale for every curve. Score = (adjusted R^2 + test R^2)/2; * marks the selected curve. k = number of parameters.

Relation	Curve	k	R^2	adj. R^2	RMSE	MAE	R^2 test	Score	Note
meantemp $\sim$ humidity	linear	2	0.329	0.329	6.009	5.113	0.122	0.225	
	quadratic	3	0.329	0.328	6.009	5.114	0.121	0.225	
	cubic *	4	0.331	0.329	6.001	5.108	0.132	0.231	
	exponential	2	0.326	0.325	6.023	5.162	0.113	0.219	
	logarithmic	2	0.313	0.313	6.079	5.245	0.086	0.200	
	power	2	0.296	0.296	6.154	5.331	0.059	0.178	
meantemp $\sim$ wind_speed	linear	2	0.098	0.098	6.966	5.991	-0.502	-0.202	
	quadratic	3	0.111	0.110	6.916	5.931	-0.603	-0.247	
	cubic	4	0.111	0.110	6.915	5.931	-0.614	-0.252	
	exponential *	2	0.094	0.093	6.982	6.014	-0.479	-0.193	
	logarithmic	2	—	—	—	—	—	—	ln(x) undefined (x <= 0)
	power	2	—	—	—	—	—	—	ln(x) undefined (x <= 0)
meantemp $\sim$ meanpressure	linear	2	0.776	0.776	3.472	2.834	0.755	0.765	
	quadratic	3	0.815	0.815	3.157	2.511	0.682	0.748	
	cubic *	4	0.819	0.818	3.124	2.478	0.715	0.767	
	exponential	2	0.730	0.730	3.808	3.160	0.716	0.723	
	logarithmic	2	0.775	0.775	3.481	2.842	0.755	0.765	
	power	2	0.729	0.729	3.819	3.170	0.715	0.722	
humidity $\sim$ wind_speed	linear	2	0.160	0.160	15.535	12.358	0.099	0.129	
	quadratic	3	0.169	0.168	15.450	12.336	0.081	0.125	
	cubic	4	0.171	0.170	15.431	12.302	0.070	0.120	
	exponential *	2	0.165	0.164	15.492	12.337	0.097	0.130	
	logarithmic	2	—	—	—	—	—	—	ln(x) undefined (x <= 0)
	power	2	—	—	—	—	—	—	ln(x) undefined (x <= 0)
humidity $\sim$ meanpressure	linear	2	0.115	0.114	15.946	12.759	0.133	0.124	
	quadratic *	3	0.187	0.186	15.287	12.212	0.292	0.239	
	cubic	4	0.187	0.185	15.287	12.212	0.283	0.234	
	exponential	2	0.123	0.122	15.875	12.689	0.152	0.137	
	logarithmic	2	0.114	0.114	15.951	12.764	0.132	0.123	
	power	2	0.122	0.122	15.880	12.694	0.151	0.136	
wind_speed $\sim$ meanpressure	linear	2	0.090	0.089	3.951	3.163	-0.356	-0.134	
	quadratic	3	0.090	0.088	3.951	3.163	-0.356	-0.134	
	cubic *	4	0.091	0.089	3.948	3.156	-0.351	-0.131	
	exponential	2	0.089	0.088	3.953	3.165	-0.355	-0.134	
	logarithmic	2	0.090	0.089	3.951	3.163	-0.356	-0.134	
	power	2	0.089	0.088	3.953	3.165	-0.355	-0.134	

Table 8: Selected curve per relation (both directions) compared with the linear fit: full-data R^2 , gain in R^2 , and R^2 on the 2017 test period.

Relation	Best curve	R^2 linear	R^2 best	Gain	R^2 test linear	R^2 test best	RMSE linear	RMSE best
meantemp $\sim$ humidity	cubic	0.329	0.331	0.002	0.122	0.132	6.009	6.001
humidity $\sim$ meantemp (rev.)	linear	0.329	0.329	0.000	0.310	0.310	13.885	13.885
meantemp $\sim$ wind_speed	exponential	0.098	0.094	-0.004	-0.502	-0.479	6.966	6.982
wind_speed $\sim$ meantemp (rev.)	logarithmic	0.098	0.089	-0.010	-0.340	-0.296	3.932	3.953
meantemp $\sim$ meanpressure	cubic	0.776	0.819	0.043	0.755	0.715	3.472	3.124
meanpressure $\sim$ meantemp (rev.)	quadratic	0.776	0.788	0.012	0.730	0.752	3.504	3.407
humidity $\sim$ wind_speed	exponential	0.160	0.165	0.005	0.099	0.097	15.535	15.492
wind_speed $\sim$ humidity (rev.)	power	0.160	0.151	-0.009	-0.004	0.044	3.795	3.815
humidity $\sim$ meanpressure	quadratic	0.115	0.187	0.072	0.133	0.292	15.946	15.287
meanpressure $\sim$ humidity (rev.)	cubic	0.115	0.125	0.010	-0.213	-0.211	6.963	6.924
wind_speed $\sim$ meanpressure	cubic	0.090	0.091	0.001	-0.356	-0.351	3.951	3.948
meanpressure $\sim$ wind_speed (rev.)	linear	0.090	0.090	0.000	-0.772	-0.772	7.063	7.063

- **Humidity–pressure:** the quadratic raises R^2 from 0.115 to 0.187 ($\Delta R^2 = +0.072$), and the gain is confirmed out of sample (test R^2 0.133 $\rightarrow$ 0.292). The U-shape is real in the data but reflects the order of the seasons along the pressure axis: humid low-pressure monsoon days, dry pre-monsoon days at intermediate pressure and humid high-pressure winter days.
- **Temperature–pressure:** the cubic removes the curvature seen in the linear residuals and raises R^2 from 0.776 to 0.819 ($\Delta R^2 = +0.043$). The test R^2 , however, falls from 0.755 to 0.715, because the curvature lies at summer temperatures that the January–April test period does not contain. The cubic also peaks at 998.0 hPa and turns down at the lowest pressures, where few days lie, so it must not be extrapolated.
- **Temperature–humidity:** the best curve (cubic) adds only +0.002 to R^2 . No function of humidity alone can separate the monsoon band from the rest.
- **Relations with wind speed:** the curves change R^2 by at most +0.013 (temperature), +0.011 (humidity) and +0.001 (pressure); a straight line is adequate. For temperature–wind speed the selected exponential even has a slightly *lower* R^2 than the line (−0.004) and was chosen only for a marginally better test R^2 .

Overfitting. For 4 of the 12 relations the cubic beats the quadratic on adjusted R^2 but loses on the test period; the selection score (Section 3.4) therefore rejects several in-sample gains, e.g. the cubic for temperature–wind speed (+0.013). The residuals of the selected curves (Figure 12) no longer show systematic curvature, but their Durbin–Watson statistics remain far below 2.

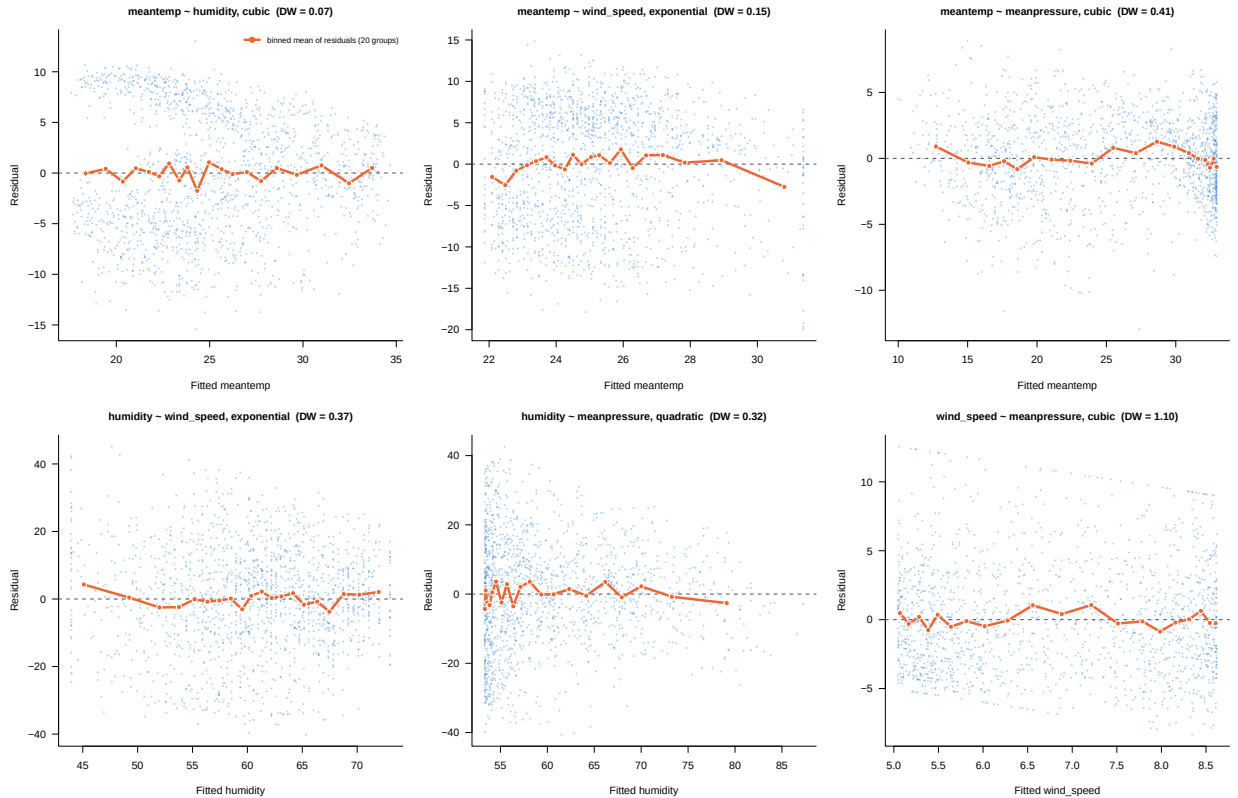


Figure 12: Residuals of the selected curve for each pair, with binned means.

8 R^2 comparison and discussion (2d)

Table 9 collects the R^2 of every curve for every relation, Figure 13 shows the linear and selected R^2 as matrices and Figure 14 the gain of the best non-linear curve over the line.

Table 9: R^2 (original scale, full data) of every curve for every relation. Selected = curve with the highest score (adjusted R^2 + test R^2)/2; ΔR^2 = best non-linear curve minus linear; – = not defined ($\ln x$ with $x \leq 0$).

Relation	Linear	Quadr.	Cubic	Exp.	Log.	Power	Selected	ΔR^2	R^2 test sel.
meantemp $\sim$ humidity	0.329	0.329	0.331	0.326	0.313	0.296	cubic	+0.002	0.132
humidity $\sim$ meantemp (rev.)	0.329	0.340	0.372	0.320	0.309	0.296	linear	-0.020	0.310
meantemp $\sim$ wind_speed	0.098	0.111	0.111	0.094	–	–	exponential	-0.004	-0.479
wind_speed $\sim$ meantemp (rev.)	0.098	0.106	0.110	0.103	0.089	0.095	logarithmic	-0.010	-0.296
meantemp $\sim$ meanpressure	0.776	0.815	0.819	0.730	0.775	0.729	cubic	+0.043	0.715
meanpressure $\sim$ meantemp (rev.)	0.776	0.788	0.796	0.775	0.719	0.717	quadratic	+0.012	0.752
humidity $\sim$ wind_speed	0.160	0.169	0.171	0.165	–	–	exponential	+0.005	0.097
wind_speed $\sim$ humidity (rev.)	0.160	0.162	0.166	0.162	0.159	0.151	power	-0.009	0.044
humidity $\sim$ meanpressure	0.115	0.187	0.187	0.123	0.114	0.122	quadratic	+0.072	0.292
meanpressure $\sim$ humidity (rev.)	0.115	0.124	0.125	0.115	0.122	0.122	cubic	+0.010	-0.211
wind_speed $\sim$ meanpressure	0.090	0.090	0.091	0.089	0.090	0.089	cubic	+0.001	-0.351
meanpressure $\sim$ wind_speed (rev.)	0.090	0.103	0.103	0.090	–	–	linear	+0.000	-0.772

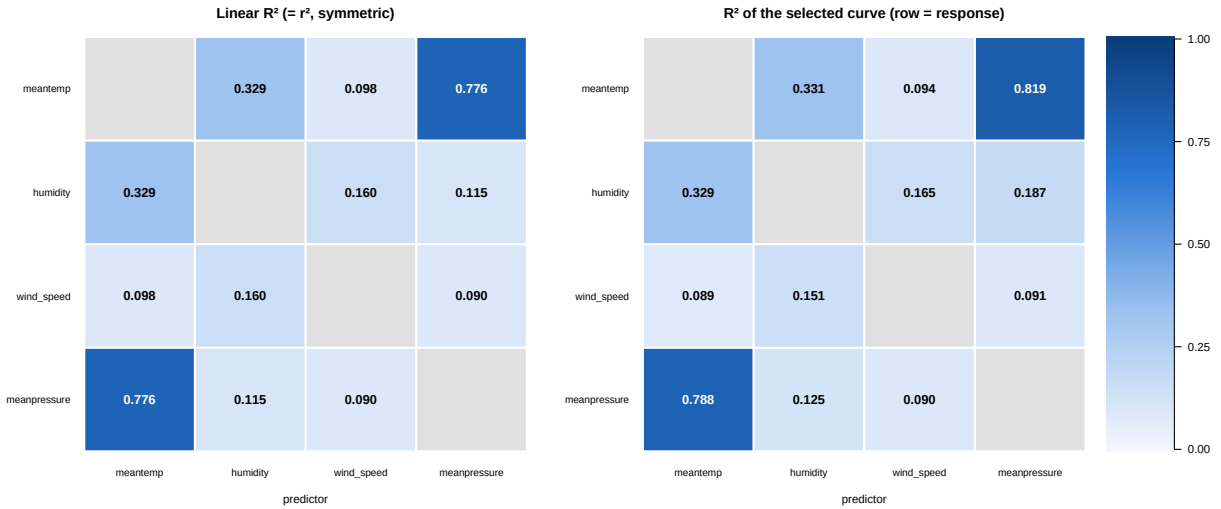


Figure 13: Linear R^2 ($= r^2$, symmetric) and R^2 of the selected curve (row = response).

Table 10 ranks the six relations by the R^2 of their selected curve and adds, in the last column, the R^2 that remains once the annual cycle is removed; the two rankings differ markedly, which the seasonality discussion below explains.

Are the high R^2 values driven by a few outliers? No. Removing the 42–80 days with Cook’s $D > 4/n$ changes the linear R^2 by +0.020 to +0.044 (Table 11): it *rises* in every case, so the influential days are poorly fitted days that lower the fit, and no reported R^2 is inflated by a few extreme observations. The wind-speed capping does matter slightly: with the raw wind speeds the linear R^2 would be 0.082, 0.139 and 0.074 instead of 0.098, 0.160 and 0.090. Part of these (weak) relations therefore depends on that pre-processing decision, although the conclusion “weak” does not change.

Seasonality. Table 12 compares the correlation of the raw data with the correlation of the anomalies (annual cycle removed) and within each season.

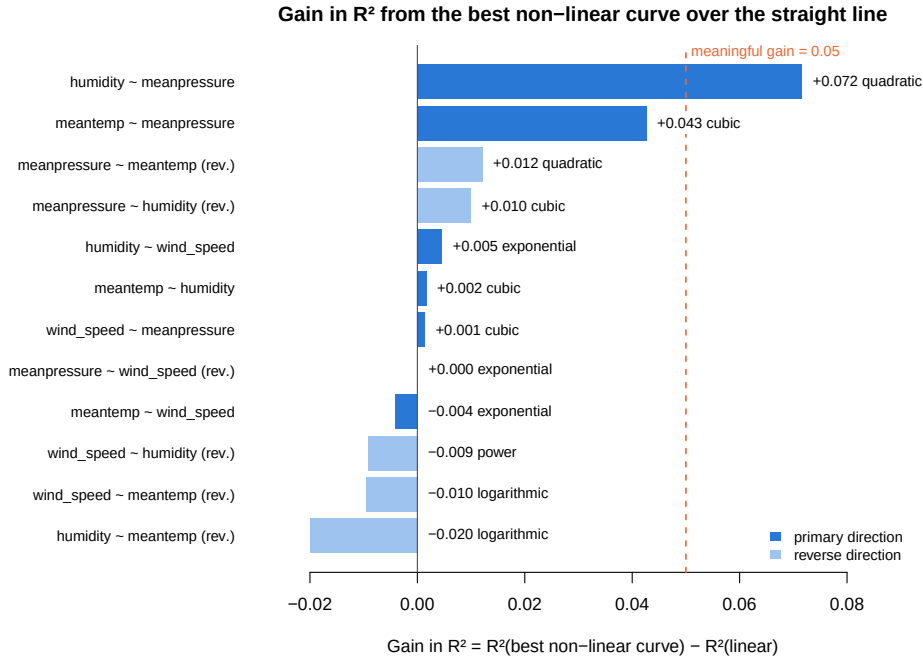


Figure 14: Gain in R^2 of the best non-linear curve over the straight line; dashed: threshold for a meaningful gain (0.05).

Table 10: Relations ranked by R^2 of the selected curve (primary direction), with the R^2 that remains after removing the annual cycle (anomaly R^2).

Rank	Relation	r	R^2 linear	Selected	R^2 selected	R^2 test	Strength	R^2 anomaly
1	meantemp ~ meanpressure	-0.881	0.776	cubic	0.819	0.715	strong	0.186
2	meantemp ~ humidity	-0.574	0.329	cubic	0.331	0.132	moderate	0.302
3	humidity ~ meanpressure	0.339	0.115	quadratic	0.187	0.292	weak	0.025
4	humidity ~ wind_speed	-0.400	0.160	exponential	0.165	0.097	weak	0.100
5	meantemp ~ wind_speed	0.313	0.098	exponential	0.094	-0.479	very weak	0.019
6	wind_speed ~ meanpressure	-0.299	0.090	cubic	0.091	-0.351	very weak	0.011

Table 11: Robustness of the linear fits: number of influential days (Cook's $D > 4/n$), R^2 after removing them, and R^2 with the uncapped wind speed.

Relation	R^2 linear	n influential	max Cooks D	R^2 without influential	Change	R^2 uncapped wind
meantemp ~ humidity	0.329	42	0.005	0.360	0.031	–
meantemp ~ wind_speed	0.098	44	0.020	0.141	0.043	0.082
meantemp ~ meanpressure	0.776	69	0.013	0.807	0.031	–
humidity ~ wind_speed	0.160	67	0.020	0.180	0.020	0.139
humidity ~ meanpressure	0.115	80	0.008	0.159	0.044	–
wind_speed ~ meanpressure	0.090	66	0.012	0.117	0.028	0.074

- **Temperature–pressure** is largely seasonal: r falls from -0.881 to -0.431 on the anomalies, so 76% of the R^2 is the shared annual cycle (hot, low-pressure summer and monsoon vs. cool, high-pressure winter). A real day-to-day link remains, strongest within the pre-monsoon season ($r = -0.86$) and weakest in the monsoon ($r = -0.31$).
- **Temperature–humidity** is *not* mainly seasonal: r changes only from -0.574 to -0.550 (8% seasonal), and within the summer and the monsoon the relation is even stronger ($r = -0.81$ and -0.80). Mixing the seasons *weakens* the overall correlation, because monsoon days are both hot and humid (on average 31.4°C and 63% humidity) and form a separate band. This is why the overall R^2 is only moderate and why no single curve in humidity improves it.
- **Humidity–pressure** (78% seasonal), **temperature–wind speed** (81%) and **wind speed–pressure** (88%) are mostly seasonal; the humidity–pressure relation is strong within the pre-monsoon season ($r = 0.70$) but absent in the monsoon ($r = 0.01$).

Table 12: Seasonality check: correlation of the raw daily values and of the anomalies from the monthly mean (annual cycle removed), the share of R^2 lost by removing the annual cycle, and the correlation within each season.

Relation	r all	R^2 all	r anomaly	R^2 anomaly	Seasonal share	r Winter	r Summer	r Monsoon	r Postmonsoon
meantemp $\sim$ humidity	-0.574	0.329	-0.550	0.302	0.081	-0.469	-0.805	-0.801	-0.160
meantemp $\sim$ wind_speed	0.313	0.098	0.137	0.019	0.808	0.190	0.137	0.301	0.215
meantemp $\sim$ meanpressure	-0.881	0.776	-0.431	0.186	0.761	-0.508	-0.864	-0.311	-0.771
humidity $\sim$ wind_speed	-0.400	0.160	-0.315	0.100	0.378	-0.315	-0.252	-0.402	-0.469
humidity $\sim$ meanpressure	0.339	0.115	0.157	0.025	0.784	0.190	0.699	0.006	0.068
wind_speed $\sim$ meanpressure	-0.299	0.090	-0.104	0.011	0.879	-0.100	-0.170	-0.161	-0.205

What the regressions do not show. Our temperature–pressure model predicts temperature *from* pressure, but that is only the direction we chose for the regression. Physically the chain runs, if anything, the other way: strong heating of north India in June goes together with the seasonal low pressure over the region, while the cool winter air mass brings high pressure. Both variables follow the same annual forcing, which is exactly why removing the monthly means (Table 12) removes most of their correlation. The same caution applies to every pair: a high R^2 here means that one parameter is a good *statistical* predictor of another in Delhi’s climate, not that changing one would change the other.

9 Conclusion

1. **Strongest relation: temperature and pressure.** $r = -0.881$; a straight line explains 0.776 of the temperature variance (slope $-0.8729^\circ\text{C/hPa}$) and the cubic 0.819. The line generalises to 2017 ($R^2_{\text{test}} = 0.755$), the cubic less well (0.715). About 76% of this R^2 is the shared annual cycle.
2. **Moderate relation: temperature and humidity.** $r = -0.574$, linear $R^2 = 0.329$; non-linear curves add nothing ($\Delta R^2 = +0.002$). This relation survives the removal of the seasons (anomaly $r = -0.550$) and is a genuine day-to-day link: hotter days are drier in every season (only weakly so in the post-monsoon season, $r = -0.16$).
3. **Weak relations:** humidity–wind speed ($R^2 = 0.160$) and humidity–pressure (0.115 linear, 0.187 quadratic). **Very weak:** temperature–wind speed (0.098) and wind speed–pressure (0.090). Wind speed is only weakly tied to the other parameters.

4. **Non-linear models help meaningfully for one relation only**, humidity and pressure ($\Delta R^2 = +0.072$, confirmed on 2017). For temperature–pressure the cubic gains $+0.043$ in-sample, but this is not confirmed out of sample. For the remaining four relations the gain is below 0.01 and the straight line is adequate.
5. **Reliability**. No R^2 is inflated by a few outliers; the wind capping raises the weak wind R^2 values by 0.015–0.021. The residuals are strongly autocorrelated (DW 0.07–1.13), so p -values are optimistic, yet all six correlations remain significant with the effective sample size. The 2017 test period covers only January–April and is therefore a limited check.

10 Individual contributions

The work was divided into four parts of comparable size, one per stage of the analysis. Each member designed and implemented the methods of their part in the common R script, produced its tables and figures and wrote the corresponding report sections. The results and conclusions (Sections 8 and 9) were discussed and agreed by the whole group. Table 13 summarises the split.

Table 13: Individual contributions (each member about 25% of the work).

Member	Responsibility (analysis and code)	Report
Aditya Palapati (group leader)	Data pre-processing (2a): merging the two files and resolving the overlapping date, detection and interpolation of invalid pressures, wind-speed outlier analysis and capping, date-continuity and range checks. Project coordination and integration of all parts into one reproducible script.	Dataset description, data pre-processing, final editing
Meghana Kuruva	Relation analysis (2b): descriptive statistics, Pearson and Spearman correlation with significance tests and confidence intervals, autocorrelation and effective sample size, seasonal profiles and the anomaly-based seasonality analysis.	Relation analysis, seasonality discussion
Rohin Sai Bogadi	Simple linear regression (2c-i): least-squares estimation, inference for the slope, train/test evaluation, residual diagnostics (Durbin–Watson, Q–Q plots, Cook’s distance). Verification of all hand-written results against R’s built-in functions.	Linear regression, robustness checks
Srimeenakshi K S	Simple non-linear regression (2c-ii): polynomial normal equations with Gaussian elimination, Gauss–Newton fitting of the exponential and power models, logarithmic model, model-selection rule and the R^2 comparison (2d).	Methodology, non-linear regression, R^2 comparison

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A Code excerpts

The excerpts below are read directly from `R/Project_code.R`; the line ranges are generated by the script, so they always show the submitted code. The full script contains six sections (setup and helpers, pre-processing, relation analysis, linear regression, non-linear regression, R^2 comparison).

Average ranks and Pearson correlation

```
1 # my_rank: ranks 1..n, ties receive the average of the positions they occupy
2 #   If sorted values  $x_{(i)} = \dots = x_{(j)}$  are tied, each gets rank  $(i + j) / 2$ .
3 #   (Spearman's rho = Pearson correlation of the ranks.)
4 my_rank <- function(x) {
5   n <- length(x)
6   o <- order(x)
7   xs <- x[o]
8   r <- numeric(n)
9   i <- 1
10  while (i <= n) {
11    j <- i
12    while (j < n && xs[j + 1] == xs[i]) j <- j + 1
13    r[o[i:j]] <- (i + j) / 2
14    i <- j + 1
15  }
16  r
17 }
18
19 # my_pearson: Pearson product-moment correlation coefficient
20 #    $S_{xx} = \sum (x_i - \bar{x})^2$ ,  $S_{yy} = \sum (y_i - \bar{y})^2$ 
21 #    $S_{xy} = \sum (x_i - \bar{x})(y_i - \bar{y})$ 
22 #    $r = S_{xy} / \sqrt{S_{xx} * S_{yy}}$ ,  $-1 \leq r \leq 1$ 
23 #   Only complete  $(x_i, y_i)$  pairs are used.
24 my_pearson <- function(x, y) {
25   ok <- !is.na(x) & !is.na(y)
26   x <- x[ok]
27   y <- y[ok]
28   dx <- x - my_mean(x)
29   dy <- y - my_mean(y)
30   sum(dx * dy) / sqrt(sum(dx^2) * sum(dy^2))
31 }
```

Simple linear regression

```
1 # my_slr: fit  $y_i = a + b * x_i + e_i$  by ordinary least squares (OLS)
2 #    $\bar{x}$ ,  $\bar{y}$  = sample means
3 #    $S_{xx} = \sum (x_i - \bar{x})^2$ 
4 #    $S_{xy} = \sum (x_i - \bar{x}) * (y_i - \bar{y})$ 
5 #    $b = S_{xy} / S_{xx}$  slope (solves  $d \text{ SSE} / d b = 0$ )
6 #    $a = \bar{y} - b * \bar{x}$  intercept (the line passes through  $(\bar{x}, \bar{y})$ )
7 #    $\hat{y}_i = a + b * x_i$  fitted values
8 #    $e_i = y_i - \hat{y}_i$  residuals
9 #    $SSE = \sum e_i^2$  error (residual) sum of squares
10 #    $SST = \sum (y_i - \bar{y})^2$  total sum of squares
11 #    $SSR = \sum (\hat{y}_i - \bar{y})^2$  regression sum of squares ( $SST = SSR + SSE$ )
12 #    $R^2 = 1 - SSE / SST$  ( $= r_{xy}^2$  in simple linear regression)
13 #    $adj \ R^2 = 1 - (1 - R^2)(n - 1)/(n - 2)$ 
14 #    $RMSE = \sqrt{SSE / n}$ ,  $MAE = (1/n) \sum |e_i|$ 
15 #    $s = \sqrt{SSE / (n - 2)}$  residual standard error (2 parameters estimated)
16 #    $SE(b) = s / \sqrt{S_{xx}}$ ,  $SE(a) = s * \sqrt{1/n + \bar{x}^2 / S_{xx}}$ 
17 #    $t_b = b / SE(b) \sim t_{\{n-2\}}$  under  $H_0$ : slope = 0,  $p = 2 * P(T_{\{n-2\}} \leq -|t_b|)$ 
18 #   95 % CI of the slope:  $b \pm t_{\{0.975, n-2\}} * SE(b)$ 
19 #   Durbin-Watson (residuals in time order):
20 #    $DW = \sum_{t=2..n} (e_t - e_{t-1})^2 / \sum_{t=1..n} e_t^2 \sim 2(1 - \rho_1)$ 
21 #    $DW \sim 2$ : no autocorrelation;  $DW < 2$ : positive autocorrelation ( $DW < 1.5$ 
22 #   is a common warning level), which makes  $SE(b)$  and  $p$ -values too small.
23 my_slr <- function(x, y) {
24   n <- length(y)
25   xbar <- my_mean(x)
26   ybar <- my_mean(y)
```

```

27 Sxx <- sum((x - xbar)^2)
28 Sxy <- sum((x - xbar) * (y - ybar))
29 b <- Sxy / Sxx
30 a <- ybar - b * xbar
31 yhat <- a + b * x
32 e <- y - yhat
33 SSE <- sum(e^2)
34 SST <- sum((y - ybar)^2)
35 SSR <- sum((yhat - ybar)^2)
36 s <- sqrt(SSE / (n - 2))
37 se_b <- s / sqrt(Sxx)
38 se_a <- s * sqrt(1 / n + xbar^2 / Sxx)
39 t_b <- b / se_b
40 tcrit <- my_qt(0.975, n - 2)
41 list(n = n, a = a, b = b, se_a = se_a, se_b = se_b, t_b = t_b,
42      p_b = 2 * pt(-abs(t_b), df = n - 2),
43      ci_b = c(b - tcrit * se_b, b + tcrit * se_b),
44      fitted = yhat, resid = e, SSE = SSE, SST = SST, SSR = SSR,
45      R2 = 1 - SSE / SST, adjR2 = adj_r2(y, yhat, n, p = 1),
46      RMSE = rmse(y, yhat), MAE = mae(y, yhat), RSE = s,
47      DW = sum((e[-1] - e[-n])^2) / sum(e^2))
48 }

```

Gaussian elimination and Gauss–Newton

```

1 # solve_linear_system: solve A beta = b (A square) by Gaussian elimination
2 # with partial pivoting; used for the least-squares normal equations
3 # (X'X) beta = X'y.
4 # Forward elimination, for column k = 1..n-1:
5 #   pivot row p = argmax_{i >= k} |A[i, k]| (largest pivot limits
6 #                                           round-off error growth)
7 #   swap rows k and p of A and b
8 #   for each row i > k: m = A[i, k] / A[k, k]
9 #                       A[i, ] <- A[i, ] - m * A[k, ]
10 #                       b[i] <- b[i] - m * b[k]
11 # Back substitution, for i = n..1:
12 #   beta_i = (b_i - sum_{j > i} A[i, j] * beta_j) / A[i, i]
13 solve_linear_system <- function(A, b) {
14   A <- as.matrix(A)
15   b <- as.numeric(b)
16   n <- nrow(A)
17   if (ncol(A) != n || length(b) != n) stop("solve_linear_system: A must be n x n and b length n")
18   tol <- 1e-12 * max(abs(A))
19   for (k in seq_len(n)) {
20     p <- (k - 1) + which.max(abs(A[k:n, k]))
21     if (abs(A[p, k]) < tol) stop("solve_linear_system: matrix is (numerically) singular")
22     if (p != k) {
23       tmp <- A[k, ]; A[k, ] <- A[p, ]; A[p, ] <- tmp
24       tmp <- b[k]; b[k] <- b[p]; b[p] <- tmp
25     }
26     if (k < n) {
27       for (i in (k + 1):n) {
28         m <- A[i, k] / A[k, k]
29         A[i, ] <- A[i, ] - m * A[k, ]
30         b[i] <- b[i] - m * b[k]
31       }
32     }
33   }
34   beta <- numeric(n)
35   for (i in n:1) {
36     s <- if (i < n) sum(A[i, (i + 1):n] * beta[(i + 1):n]) else 0
37     beta[i] <- (b[i] - s) / A[i, i]
38   }
39   beta
40 }
41
42
43 # gauss_newton: non-linear least squares, minimise SSE(theta) = sum (y_i - f(x_i; theta))^2
44 # Linearise f around the current estimate: f(x; theta + delta) ~ f(x; theta) + J delta,
45 # with Jacobian J_ij = d f(x_i; theta) / d theta_j. Least squares for delta
46 # gives the normal equations
47 # (J'J) delta = J'r,      r = y - f(x; theta) (solved by solve_linear_system)

```

```

48 # Step halving: try theta + lambda * delta for lambda = 1, 1/2, 1/4, ...
49 # (at most 30 halvings) and accept the first step that lowers SSE, so SSE
50 # decreases monotonically.
51 # Converged when the relative decrease of SSE or the relative step size
52 # max_j |lambda delta_j| / (|theta_j| + 1e-8) falls below tol. Not converged
53 # after max_iter iterations, or when no step size lowers SSE while the step
54 # is still large. f and jac are functions of (x, theta).
55 gauss_newton <- function(f, jac, theta, x, y, max_iter = 200, tol = 1e-10) {
56   # SSE(theta) = sum (y_i - f(x_i; theta))^2; Inf if not finite (rejects the step)
57   sse <- function(th) {
58     v <- sum((y - f(x, th))^2)
59     if (is.finite(v)) v else Inf
60   }
61   S <- sse(theta)
62   if (!is.finite(S)) stop("SSE is not finite at the start values")
63   for (iter in 1:max_iter) {
64     r <- y - f(x, theta)
65     J <- jac(x, theta)
66     delta <- solve_linear_system(t(J) %*% J, t(J) %*% r)
67     lambda <- 1
68     accepted <- FALSE
69     for (h in 0:30) {
70       theta_new <- theta + lambda * delta
71       S_new <- sse(theta_new)
72       if (S_new < S) { accepted <- TRUE; break }
73       lambda <- lambda / 2
74     }
75     rel_step <- max(abs(lambda * delta) / (abs(theta) + 1e-8))
76     if (!accepted) {
77       # no improvement possible: fine if we are already at the minimum
78       ok <- max(abs(delta) / (abs(theta) + 1e-8)) < 1e-6
79       return(list(theta = theta, SSE = S, iterations = iter, converged = ok,
80                 message = if (ok) "converged" else "step halving failed"))
81     }
82     rel_dec <- (S - S_new) / S
83     theta <- theta_new
84     S <- S_new
85     if (rel_dec < tol || rel_step < tol) {
86       return(list(theta = theta, SSE = S, iterations = iter, converged = TRUE,
87                 message = "converged"))
88     }
89   }
90   list(theta = theta, SSE = S, iterations = max_iter, converged = FALSE,
91        message = sprintf("no convergence in %d iterations", max_iter))
92 }

```

Polynomial and exponential models

```

1 # fit_poly_model: polynomial least squares through the normal equations
2 # minimise ||y - X beta||^2 => (X'X) beta = X'y
3 # solved with our solve_linear_system (Gaussian elimination, partial pivoting).
4 # Standardising x keeps X'X well conditioned: with raw pressure (~1000 hPa)
5 # the columns would be ~1, 1e3, 1e6, 1e9 and X'X numerically singular.
6 # The fitted curve is the same as with raw x; the coefficients refer to u.
7 fit_poly_model <- function(x, y, d) {
8   m <- my_mean(x)
9   s <- my_sd(x)
10  X <- poly_design(x, m, s, d)
11  beta <- solve_linear_system(t(X) %*% X, t(X) %*% y)
12  names(beta) <- paste0("c", 0:d)
13  eq <- paste0("y = ", fmt_coef(beta[1]),
14             paste(sapply(1:d, function(j) term(beta[j + 1], if (j == 1) " u" else paste0(" u^", j))),
15             collapse = ""),
16             sprintf(", u = (x - %s) / %s", fmt_coef(m), fmt_coef(s)))
17  list(model = paste0("poly", d), k = d + 1, valid = TRUE, converged = TRUE, iterations = 0,
18       status = "ok (normal equations)", params = beta, center = m, scale = s, equation = eq,
19       curve = function(xn) as.vector(poly_design(xn, m, s, d) %*% beta))
20 }
21
22 # fit_exp_model: y = a * exp(b x)
23 # Computed in the equivalent centred form y = A exp(b (x - c)), c = xbar,
24 # a = A exp(-b c), so that A and b have similar magnitudes (with x ~ 1000 hPa,

```

```

25 # a itself can be ~1e15, which makes Gauss-Newton badly conditioned).
26 # 1) Start values by linearising:  $\ln y = \ln A + b (x - c)$  -> our my_slr:
27 #  $A0 = \exp(\text{intercept})$ ,  $b0 = \text{slope}$ . Requires  $y > 0$ ; if some  $y = 0$  (wind
28 # speed), the start uses only the days with  $y > 0$ .
29 # 2) Refine on the ORIGINAL scale with Gauss-Newton, Jacobian
30 #  $d f / d A = \exp(b (x - c))$ ,  $d f / d b = A (x - c) \exp(b (x - c))$ .
31 fit_exp_model <- function(x, y) {
32   pos <- y > 0
33   if (sum(pos) < 3) return(invalid_fit("exponential", "skipped: fewer than 3 values with y > 0"))
34   c0 <- my_mean(x)
35   start <- my_slr(x[pos] - c0, log(y[pos]))
36   # model  $f(x) = A \exp(b (x - c))$  and its Jacobian  $[df/dA, df/db]$ 
37   f <- function(x, th) th[1] * exp(th[2] * (x - c0))
38   jac <- function(x, th) {
39     e <- exp(th[2] * (x - c0))
40     cbind(e, th[1] * (x - c0) * e)
41   }
42   gn <- gauss_newton(f, jac, c(exp(start$a), start$b), x, y)
43   A <- gn$theta[1]
44   b <- gn$theta[2]
45   a <- A * exp(-b * c0)
46   note <- if (all(pos)) "" else sprintf("; start values from the %d values with y > 0", sum(pos))
47   list(model = "exponential", k = 2, valid = TRUE, converged = gn$converged,
48        iterations = gn$iterations,
49        status = sprintf("%s (Gauss-Newton, %d iterations)%s",
50                          if (gn$converged) "ok" else gn$message, gn$iterations, note),
51        params = c(a = a, b = b), center = c0, scale = NA,
52        equation = sprintf("y = %s * exp(%s x)", fmt_coef(a), fmt_coef(b)),
53        curve = function(xn) A * exp(b * (xn - c0)))
54 }

```